Administrator Manual

Setting up and running exams with ExamProctor

For the person who owns the system: you create students, exams and teacher accounts, hand out passwords, and look after the server. Teachers and students have their own manuals.

ExamProctor · Version 1.0
October 2026

Contents

Work through sections 1 to 5 once to get ready. After that you will mostly use sections 7 and 8 on exam days.

1 How the system fits together

2 Before you start

3 A tour of the admin site

4 Students

5 Exams

6 Teachers

7 While an exam is running

8 After the exam

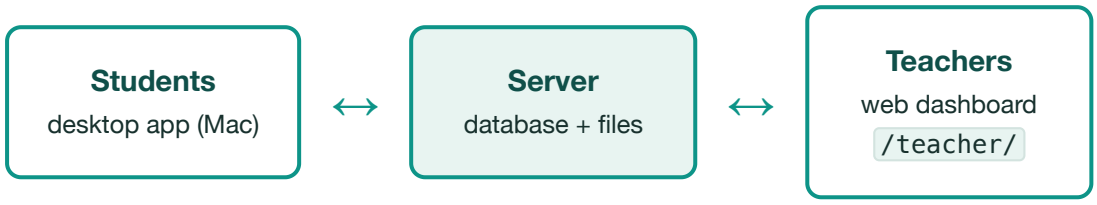
9 Server and maintenance

10 Troubleshooting

11 Quick reference

1 How the system fits together

ExamProctor runs written exams (a Google Form) and practical exams (students work in their own apps) while quietly recording evidence: screenshots, optional webcam photos and a log of events such as switching windows. Everything is stored on **one server** that your school runs. Three groups of people use it:



You, the administrator, use the admin site at `/admin/`. Teachers have a separate, simpler dashboard with their own login and never see the admin site.

Words you will see everywhere

Word	Meaning
Batch	A number on every student (for example 81). **Every exam belongs to exactly one batch**, and students only see exams of their own batch. A student's batch can only move forward (81 to 82), never back, except as a recorded correction.
Exam	Has a **written part** (Google Form), a **practical part**, or both. If both, the order is always written first, then practical.
Attempt	One per student per exam. It ends when the student finishes, quits, runs out of time, or a teacher ends it. A student who quits cannot re-enter unless a teacher or you reset the attempt.
Session	A student's login. Only one session per student at a time: logging in somewhere else ends the first one.
Evidence	Screenshots and webcam photos. They use the most disk space and are the most private, so they can be deleted after an exam while the event log and marks are kept.
Flag	A traffic light per student (clean, review, flagged) calculated from the events, to show teachers who to look at first.

2 Before you start

Addresses you need

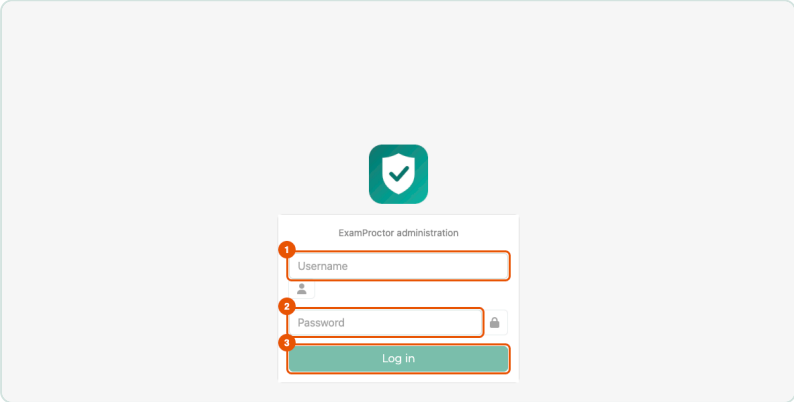
What	Address	Used by
Admin site	`http://SERVER:8000/admin/`	You
Teacher dashboard	`http://SERVER:8000/teacher/`	Teachers (and you, with your admin login)
Server home page	`http://SERVER:8000/`	Everyone (links and the installer download)

Replace `SERVER` with your server's address, for example `192.168.0.100` when students are on the same network. The student app has this address built in when the installer is made, so decide it first and keep it stable (ask the router to always give the server the same address).

> **Keep it running**
> Keep the server computer awake and connected for the whole exam. If it sleeps, students go offline; they recover when it wakes (their app keeps everything and sends it later), but teachers cannot monitor in the meantime.

Logging in

1. Open the admin address in a browser.
2. Type your **username** and **password** and press **Log in**.



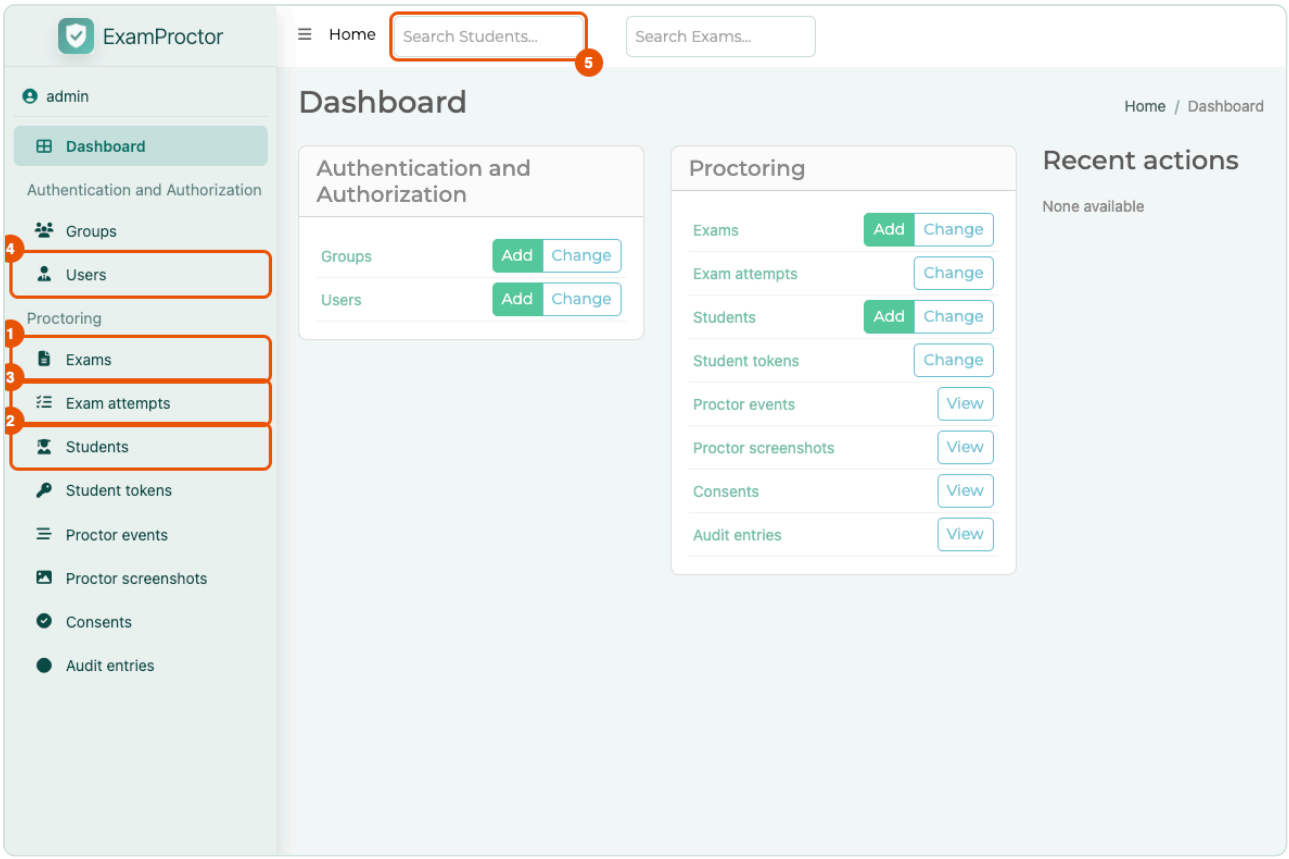
The admin login page

1. Username
2. Password
3. Log in

The first administrator account is created once on the server with `python manage.py createsuperuser`. To add more administrators, create a user in **Users** and tick **Superuser status**.

3 A tour of the admin site

The left menu is your map. Everything you do starts from here.



The admin home page

1. **Exams**: create and edit exams, archive them.
2. **Students**: add students, change batches, generate passwords, deactivate.
3. **Exam attempts**: who is taking each exam; force-end or reset an attempt.
4. **Users**: administrator and teacher accounts.
5. Quick search for students and exams.

Menu item	What it is for
Exams	The exams: parts, batch, teachers, schedule, monitoring and webcam settings, archive.
Exam attempts	One row per student per exam, with part, status and last-seen time.
Students	Student records, batches, activation, passwords. Includes the batch history of each student.
Student tokens	Active student **sessions**. Deleting one logs that student out.

Menu item	What it is for
Proctor events / screenshots / consents	The evidence. Read-only: nobody edits it by hand.
Audit entries	A permanent record of archives and evidence deletions (who and when).
Groups, Users	Accounts. Teachers are users in the group **Teachers**.

> **Good to know**
> As an administrator you can also open `/teacher/` with the same login and see every exam's live monitor. The two sites deliberately have no links to each other.

4 Students

Adding students

For a class, use a CSV file; for one or two students, use the form.

A. Import a CSV (recommended)

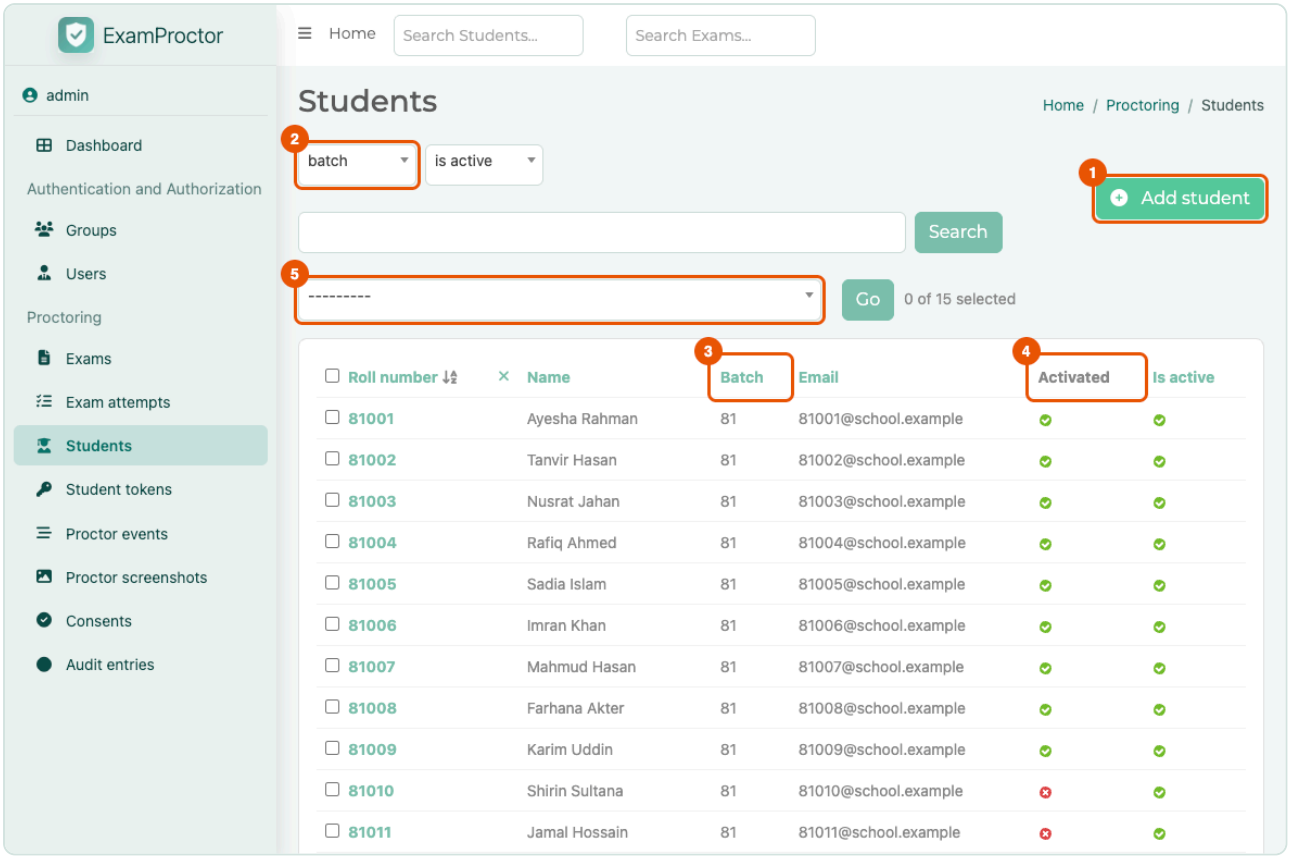
Create a file with these columns (the batch column is optional but you will want it):

```
roll_number,name,email,batch
81001,Ayesha Rahman,ayesha@school.example,81
81002,Tanvir Hasan,tanvir@school.example,81
```

```
cd backend
source .venv/bin/activate
python manage.py import_students students.csv
```

- Running it again **updates** students instead of duplicating them.
- If a row would move a student's batch **backwards**, that row is skipped and reported; nothing else is affected.
- The email is needed only for self-activation by email (section 4.3).

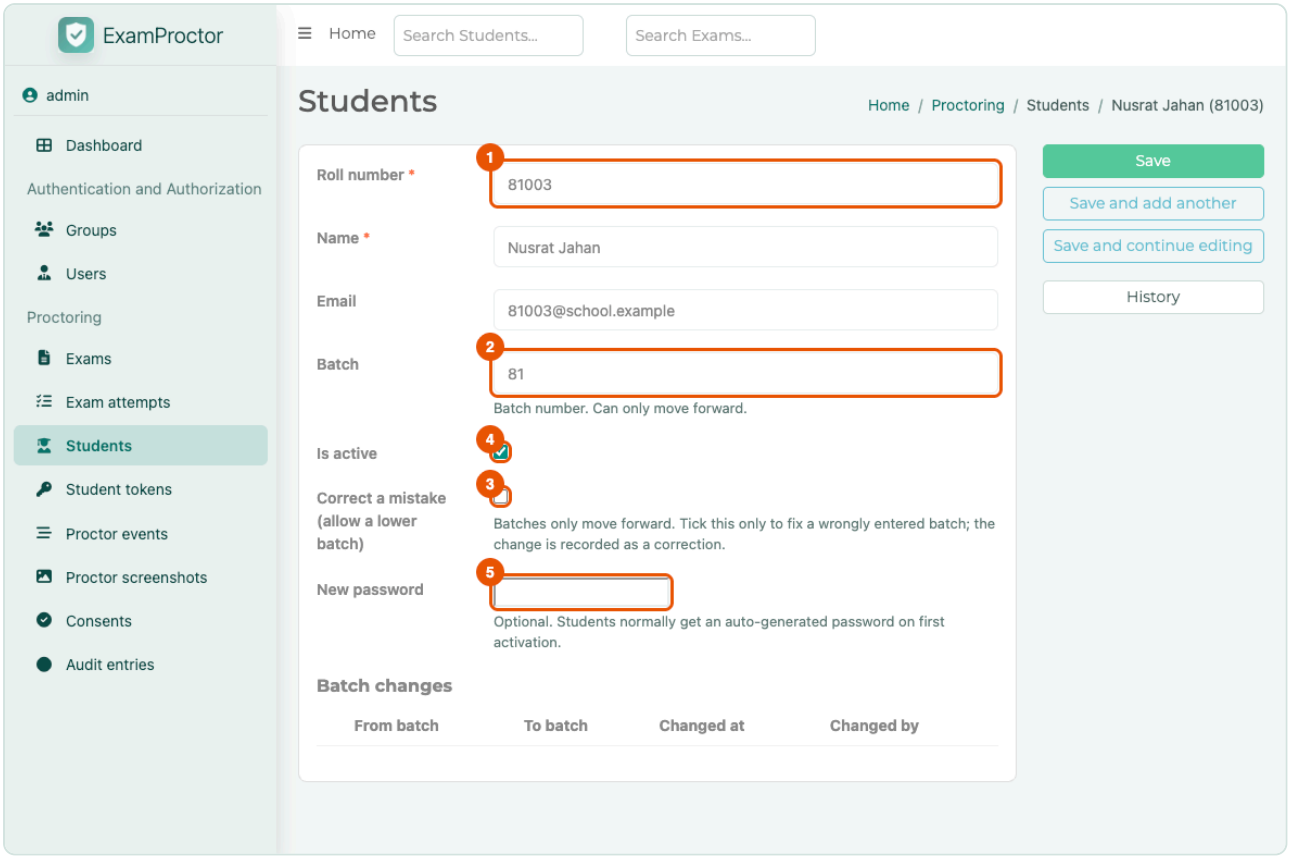
B. Add one student in the admin



The Students list

1. **Add student**.
2. Filters (by batch, by active).
3. Sort and read the **Batch** column.
4. **Activated**: a green tick means the student already has a password.
5. The **action menu**: choose an action, tick students, press Go.

The student form



Editing a student

1. Roll number (must be unique; any characters are allowed, for example 2021/001).
2. **Batch**: can move forward only.
3. **Correct a mistake**: tick only to lower a wrongly entered batch. It is recorded as a correction.
4. **Is active**: untick to stop a student logging in.
5. **New password**: optional; normally leave it empty.

Batches move forward only

A student can go from batch 81 to 82 or to 85, but not from 95 back to 94. Every move is recorded at the bottom of the student's page (who, when, from, to). If you simply typed the wrong number, tick **Correct a mistake** once; it is logged as a correction.

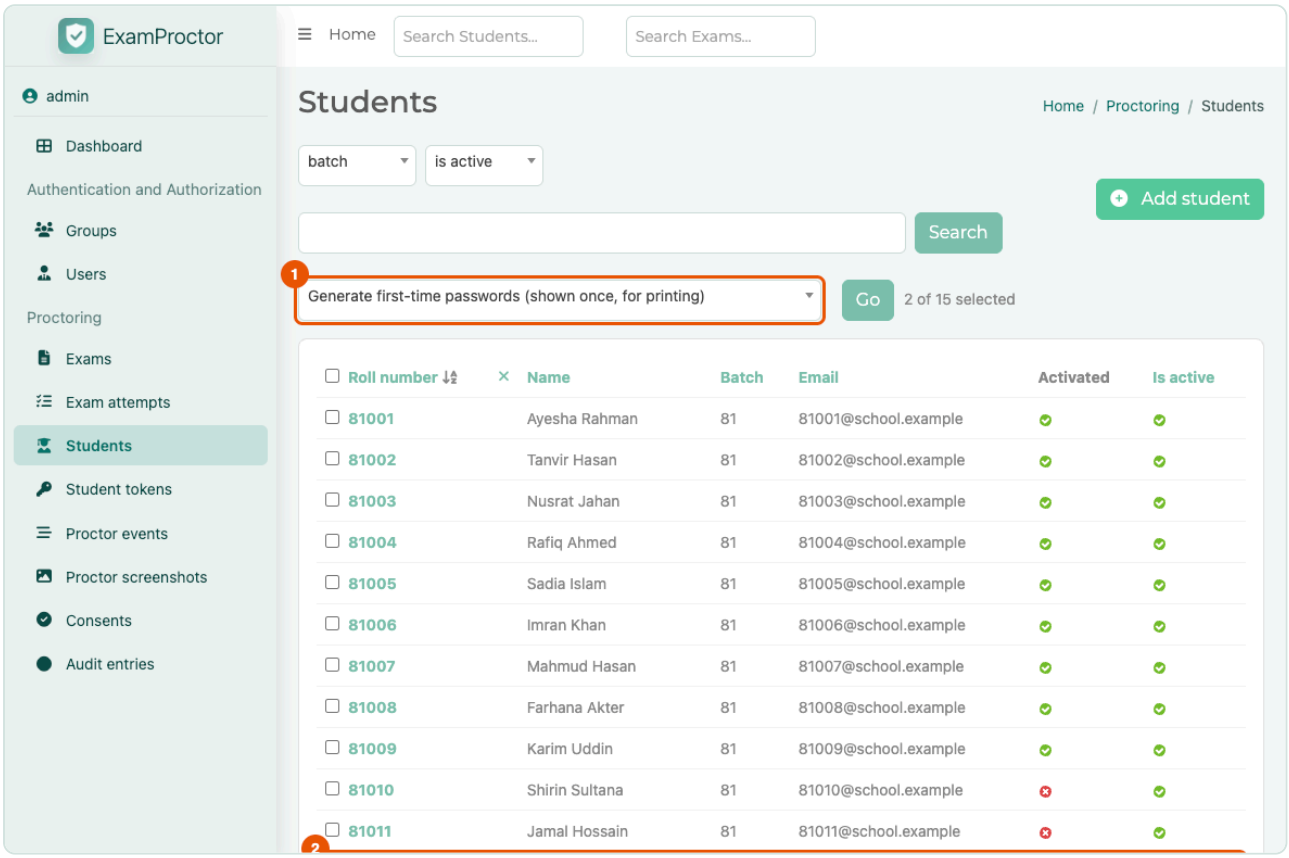
Passwords: three ways

Way	How it works	Needs
1. Email (self-activation)	The student taps *First time? Activate account* in the app, enters roll number and the email registered in the system, and the password is emailed to that address. Only the owner of the inbox gets it.	Email set up on the server (section 9)
2. Hand-out sheet	You (or a teacher) generate random passwords for students and print the one-time sheet. Best for a classroom.	Nothing

Way	How it works	Needs
3. Reset	Clears a password so the student can activate again, or generate a new one with a hand-out.	Nothing

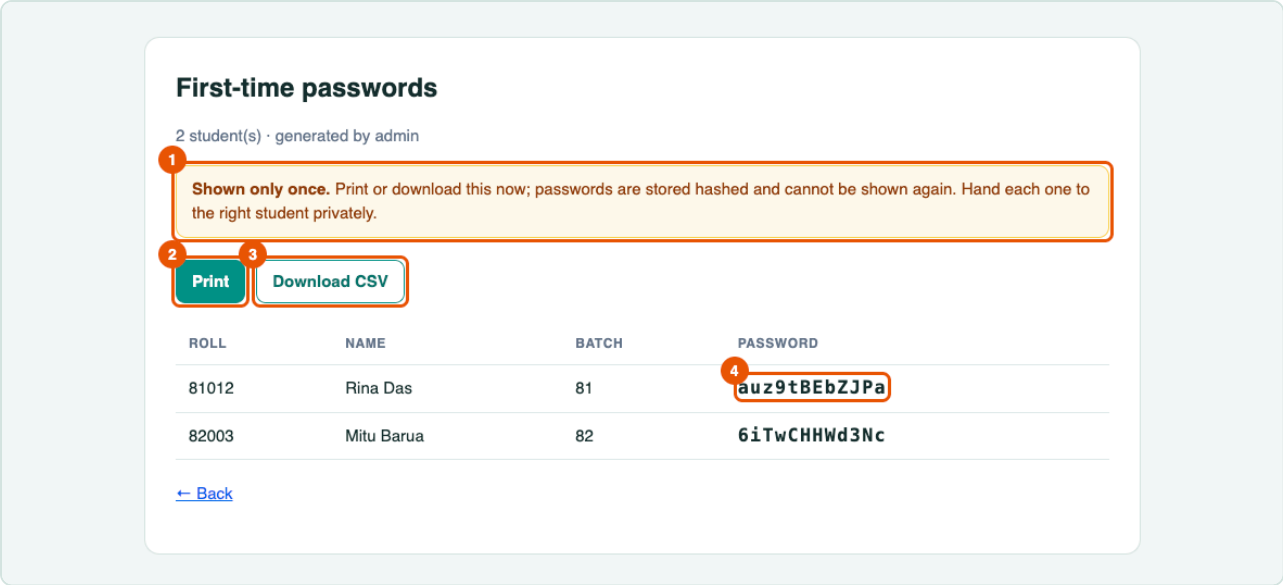
Generate hand-out passwords

1. In **Students**, tick the students.
2. Open the action menu and choose **Generate first-time passwords**, then press **Go**.
3. The one-time sheet opens. Press **Print** or **Download CSV** *now*.



Choosing the action

1. The action menu
2. The ticked students



The one-time password sheet

1. The warning: it is shown only once
2. Print
3. Download CSV (opens correctly in Excel, including Bangla names)
4. Each student's own random password

> **Shown only once**
> Passwords are stored only as hashes, so this sheet cannot be shown again. If a student loses theirs, generate a new one (their old password stops working and any current session ends). Give each student **their own** password, never one shared password for a batch.

Deactivating and deleting

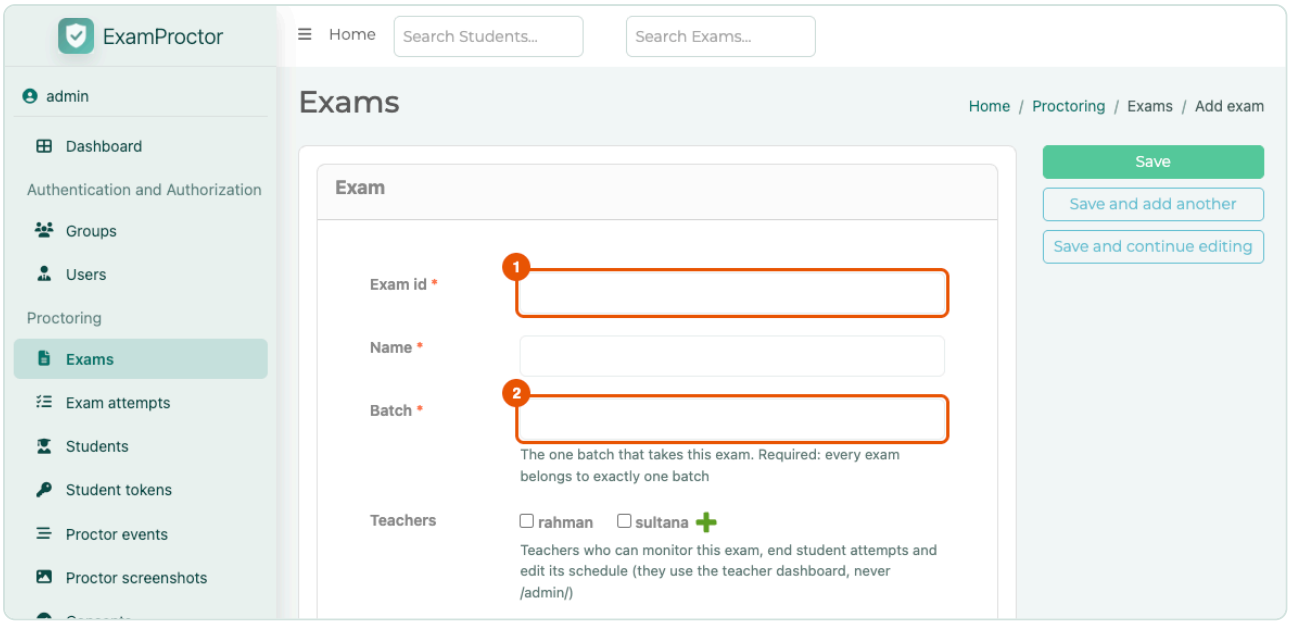
- **Deactivate** selected students (action) or a whole batch: `python manage.py deactivate_batch 81`. They can no longer log in and their sessions end, but all records are kept. Use **Reactivate** to undo.
- A student must be deactivated before you can delete the record. Deleting also deletes their attempts, so do it only after the retention period.

5 Exams

Create an exam

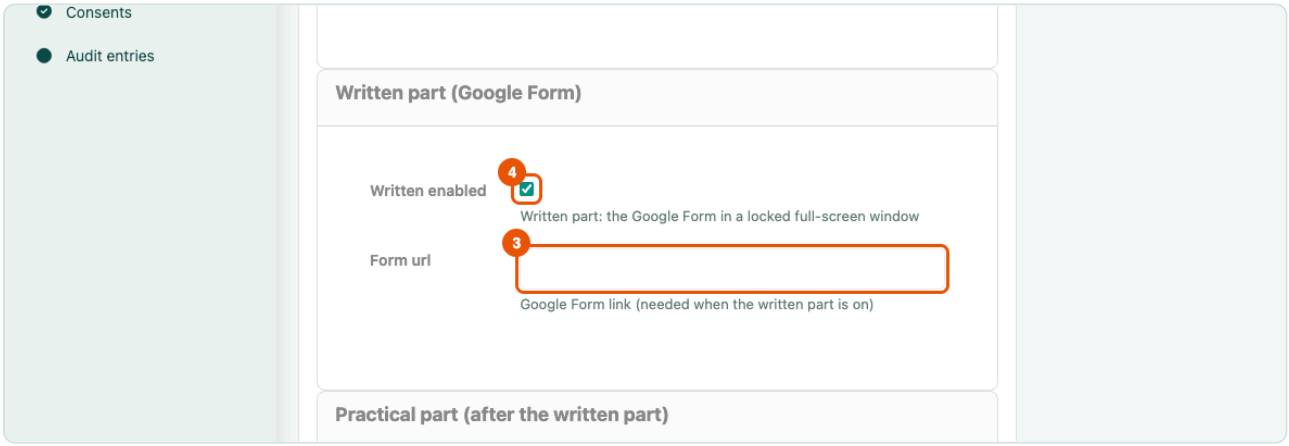
1. Open **Exams** and press **Add exam**.
2. Fill in the form (below) and press **Save**.

3. Tell the assigned teachers; they can adjust schedule and settings themselves from their dashboard.



The exam form (top)

1. **Exam id**: a short code with no spaces, for example `physics-mid`.
2. **Batch** (required): the only batch that can take this exam.
3. **Form url**: the Google Form link (needed when the written part is on).
4. **Written enabled**: the Google Form part.



The exam form (practical part)

5. **Practical enabled**: the practical part, always after the written part.
6. **Practical instructions**: shown to students in the practical window.

Every setting explained

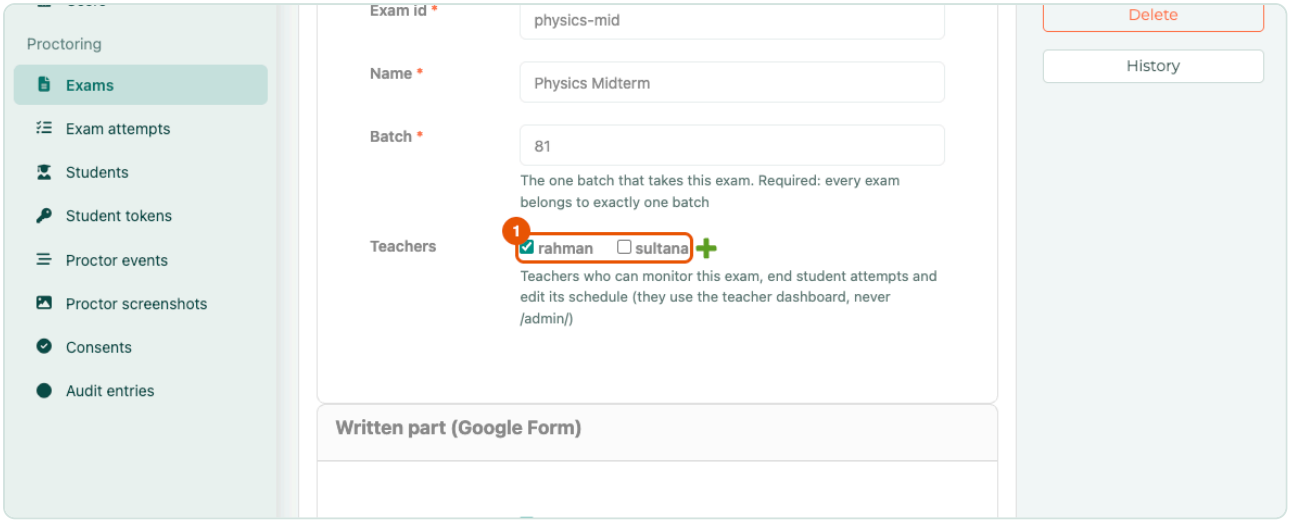
Setting	What it does
Exam id / Name	Id is the short code; Name is what everyone reads.

Setting	What it does
Batch	Required. Students outside this batch cannot see or start the exam.
Teachers	Teachers who can monitor and manage this exam (tick the boxes).
Written enabled + Form url	The Google Form opens in a locked full-screen window. Switching away, copying, developer tools and leaving the form site are blocked or logged.
Practical enabled	After the written part (or alone if written is off) students get a normal window with the timer and instructions and can use any apps. The app takes screenshots on a timer.
Practical instructions / url	Text shown to students, plus an optional link (for example a lab sheet).
Practical screenshot seconds	Average gap between screenshots in the practical part (15 to 600, default 60; randomised by up to 50 %).
Starts at / Ends at	The window in which students can start. At the end time every student's window closes with a time-up message. Leave empty for no limit.
Monitoring enabled	Master switch for the proctoring rules (screenshots, restrictions, app checks). Off means students just take the exam.
Grace seconds	Switching away for fewer seconds than this is ignored (default 3).
Webcam enabled / interval	Requires a working camera at the start and takes a photo about every N seconds (30 to 3600, default 180, randomised).
Archived	Finished and published: hidden from students and the default teacher list.

Time zone

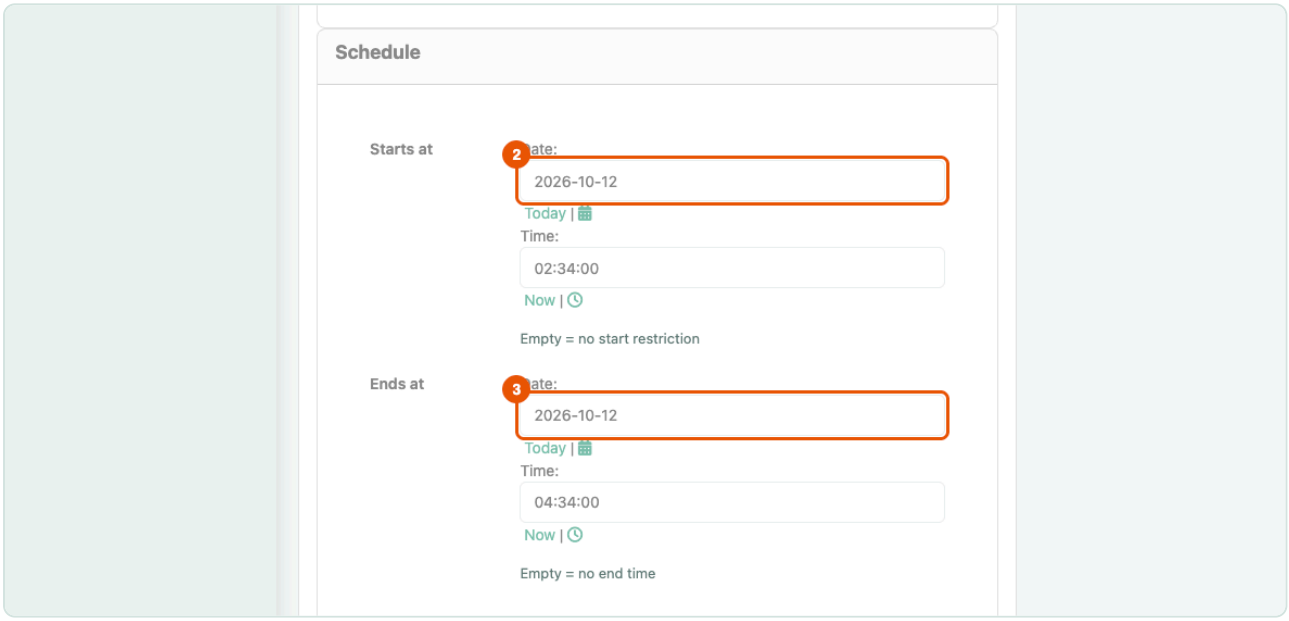
Times use the server's time zone (set with the `TIME_ZONE` setting, for example Asia/Dhaka). Check it once, then enter times as you read them on the wall clock.

Assign teachers



Choosing the exam's teachers

1. Tick the teachers for this exam.



Schedule (further down the same form)

2. **Starts at**.
3. **Ends at**.

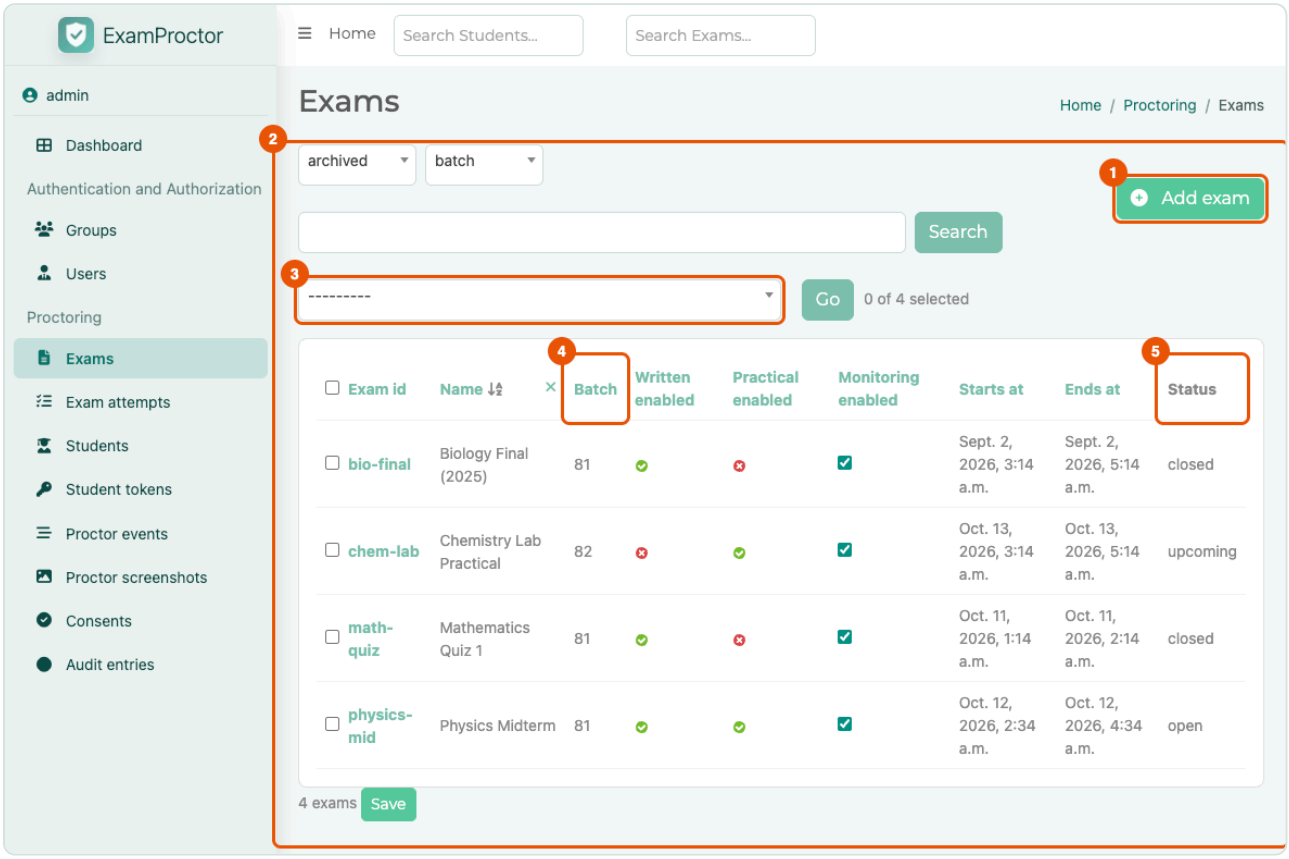
Only users in the group **Teachers** appear in this list. Create them as described in section 6.

Prepare the Google Form

- Add a question for the **roll number** (required). The marks page matches students by it.
- For automatic marks, make it a **Quiz** with answers and points so the exported CSV has a *Score* column.

- Turn on **Require sign-in** (and limit to 1 response). Then Google keeps a student's progress as a draft if the computer crashes, and they continue where they stopped. Test this once with a sample form.
- Use the link that ends in `/viewform`.

The exam list



Exams

1. **Add exam**.
2. Filter archived or active exams.
3. Actions (for example archive).
4. Batch of each exam.
5. Status: open, upcoming or closed.

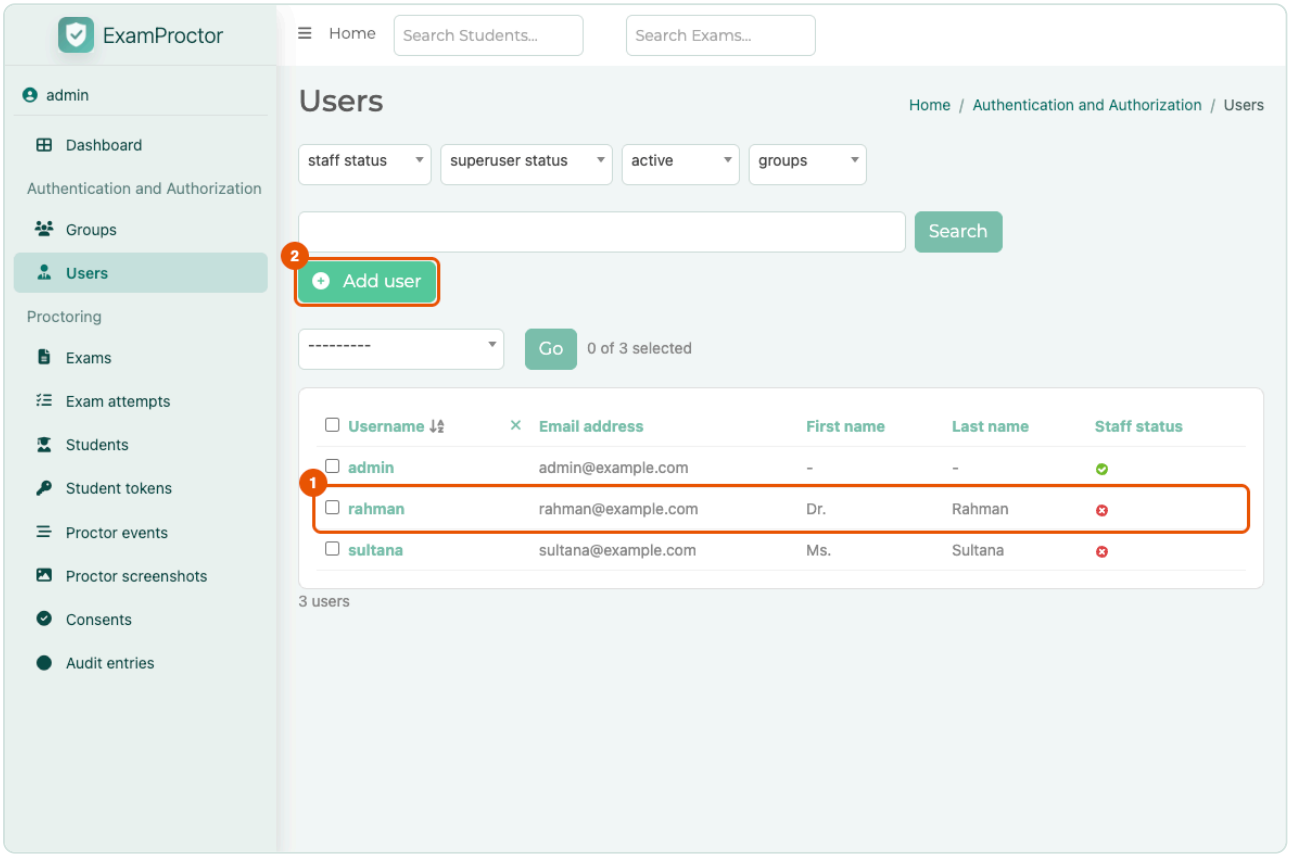
Monitoring enabled can be switched directly in the list. The change reaches students within about two minutes.

6 Teachers

Create a teacher account

```
cd backend
source .venv/bin/activate
python manage.py create_teacher rahman --exams physics-mid,math-quiz --email
rahman@school.example
```

- The command prints a one-time password for the new teacher. Give it to them privately.
- Run it again with more `--exams` to give a teacher more exams, or `--password` to set a new password.
- Teachers are **not staff**: they cannot log in to the admin site. They use `/teacher/login/`.



Users: administrators and teachers

1. A teacher account (not staff, not superuser)
2. Add user

What teachers can and cannot do

Teachers can

Only you can

- Create, delete or re-batch exams

- Watch the live monitor of **their own** exams
- Read timelines, screenshots and webcam photos
- Give extra time, end or reset an attempt, move a student to the practical part
- Generate hand-out passwords for their exam's batch or one student
- Edit their exam's schedule, parts, monitoring and webcam settings
- Upload Google Form results to get a marks sheet, download CSVs
- Archive an exam and delete its screenshots

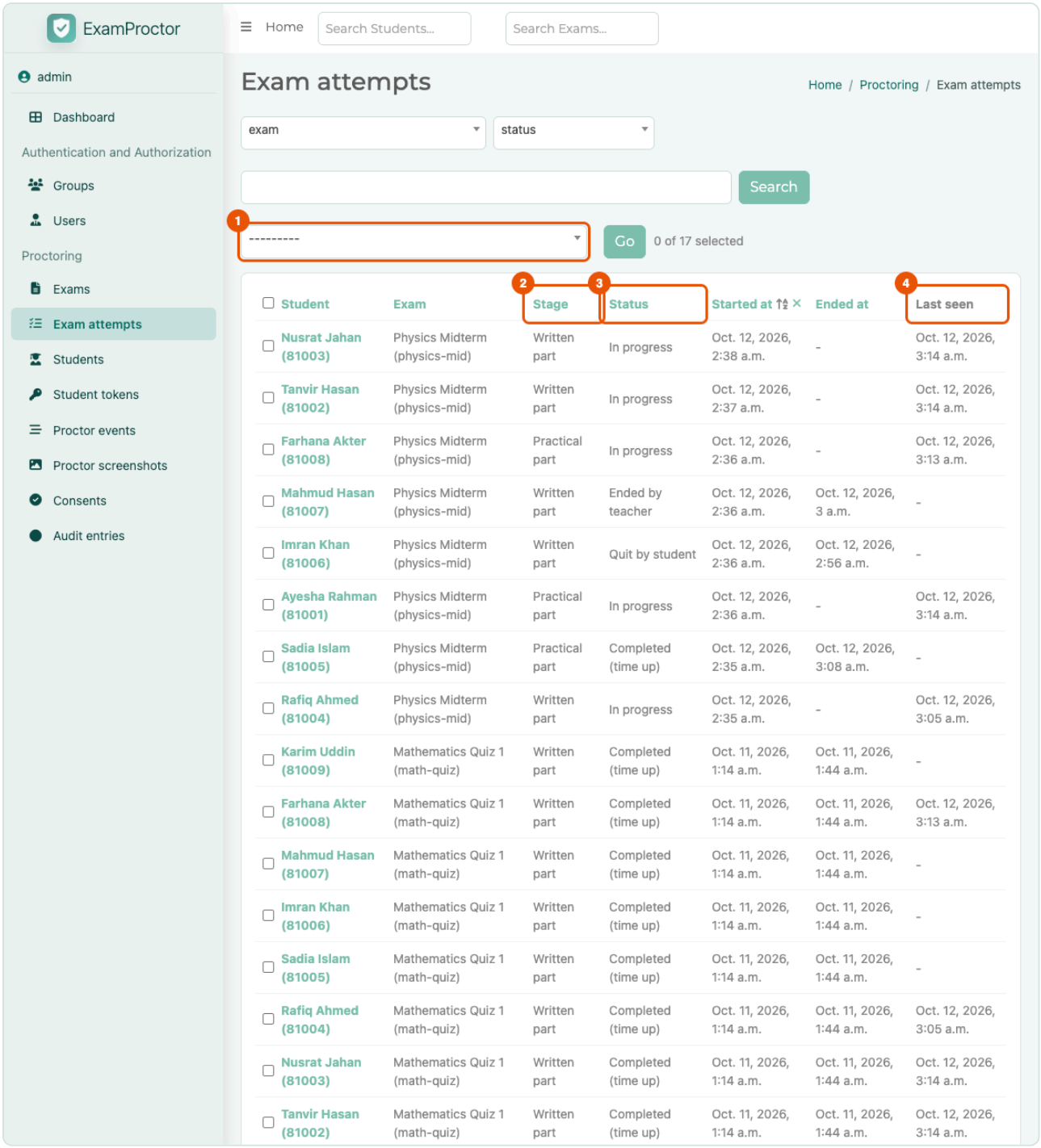
- Choose which teachers an exam has
- Manage students, batches and deactivation
- Delete users, revoke sessions, read audit entries
- Change server settings and run commands

> **Teacher cannot log in**
> If a teacher cannot log in: check they are in the group **Teachers** and not locked out (five wrong passwords lock a username for 15 minutes). To reset a password, open the user in **Users** and use the password link.

7 While an exam is running

Teachers run the exam day to day. You step in for things only an administrator can do.

Exam attempts

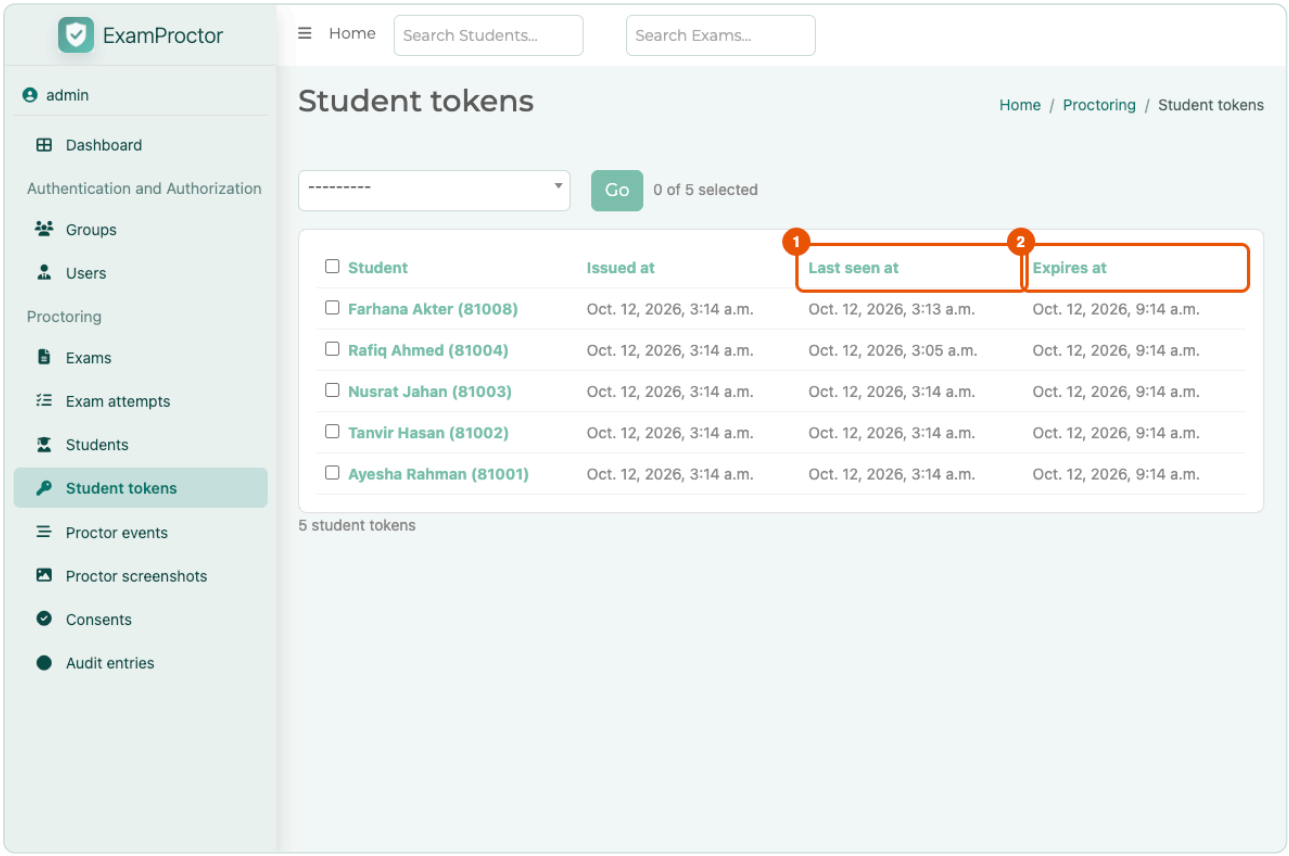


Exam attempts

1. The action menu: **Force end** or **Reset** the selected attempts.
2. **Stage**: written or practical.
3. **Status**: in progress, completed, quit or ended by teacher.
4. **Last seen**: the last heartbeat from the student's app.

Action	Effect
Force end	Ends the attempt, closes the student's exam window within about 30 seconds, blocks re-entry and logs who did it.
Reset	Deletes the attempt so the student can start again (after a technical problem or a mistake). Their earlier events stay in the log.

Sessions



Student tokens = active sessions

1. When the student was last seen
2. When the session expires (8 hours)

Delete a row to log that student out immediately. Their exam window shows *Your session has ended* and closes at its next check (within about 30 seconds).

Common requests during an exam

Request	What to do
"I forgot my password"	Students & passwords page (teacher) or **Generate first-time passwords** (you). The old password stops working.
"I can't see my exam"	Check the student's batch matches the exam's batch, the exam is not archived, and it has a batch.

Request	What to do
"I quit by mistake"	Reset the attempt. Quitting is final until someone resets it.
"My computer crashed"	Nothing to do: they log in again and continue in the same part. The restart is recorded.
Student not in the system	Add them (section 4); give them a password; make sure the batch matches.

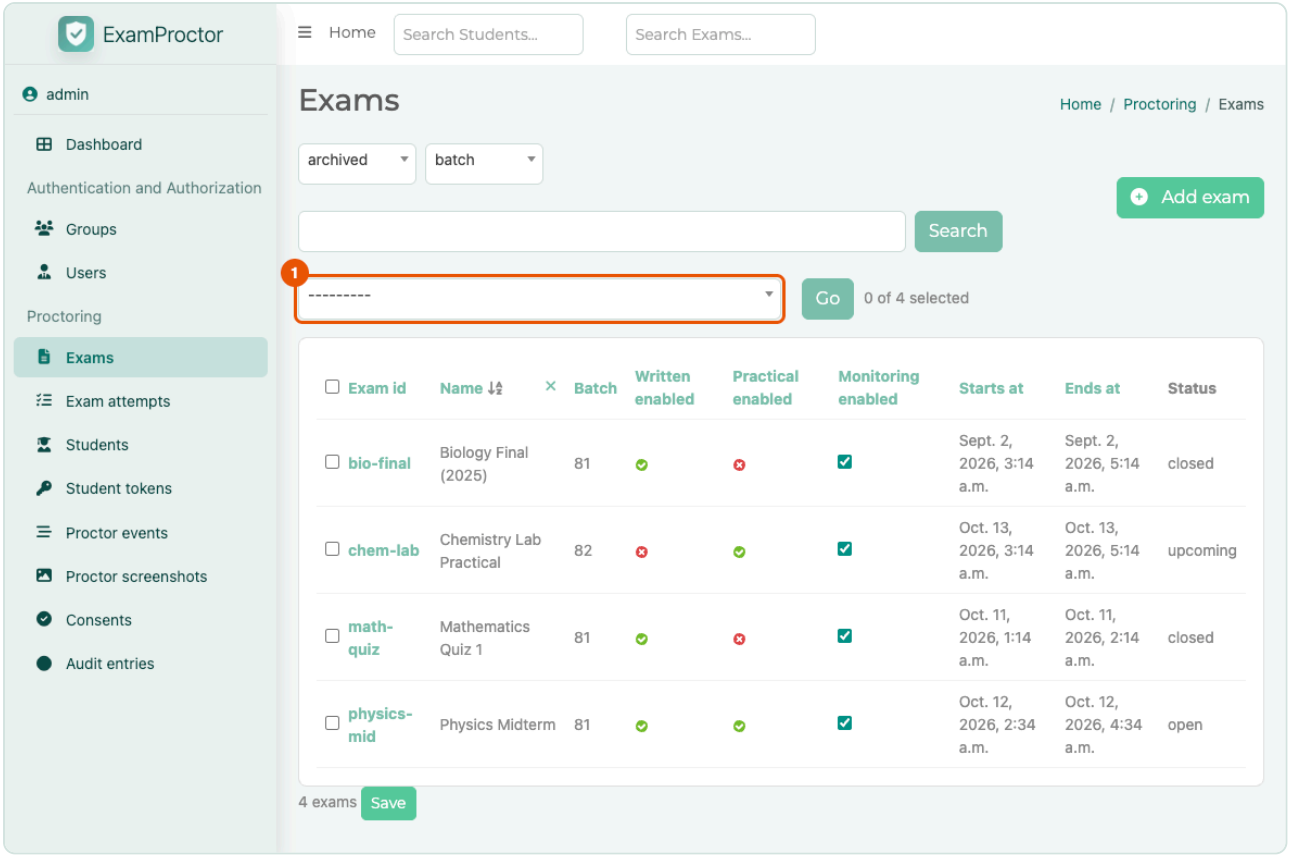
8 After the exam

Suggested routine

1. The teacher finishes marking and downloads the CSVs.
2. **Archive** the exam: it disappears from students and the default lists but stays readable.
3. After your retention period (for example 90 days) **delete the screenshots and photos**. The event log and marks stay.
4. **Deactivate** the batch when it has graduated.

Archive

Teachers archive from their dashboard; you can archive from the exam list action menu.



Exam list: action menu with Archive selected exams

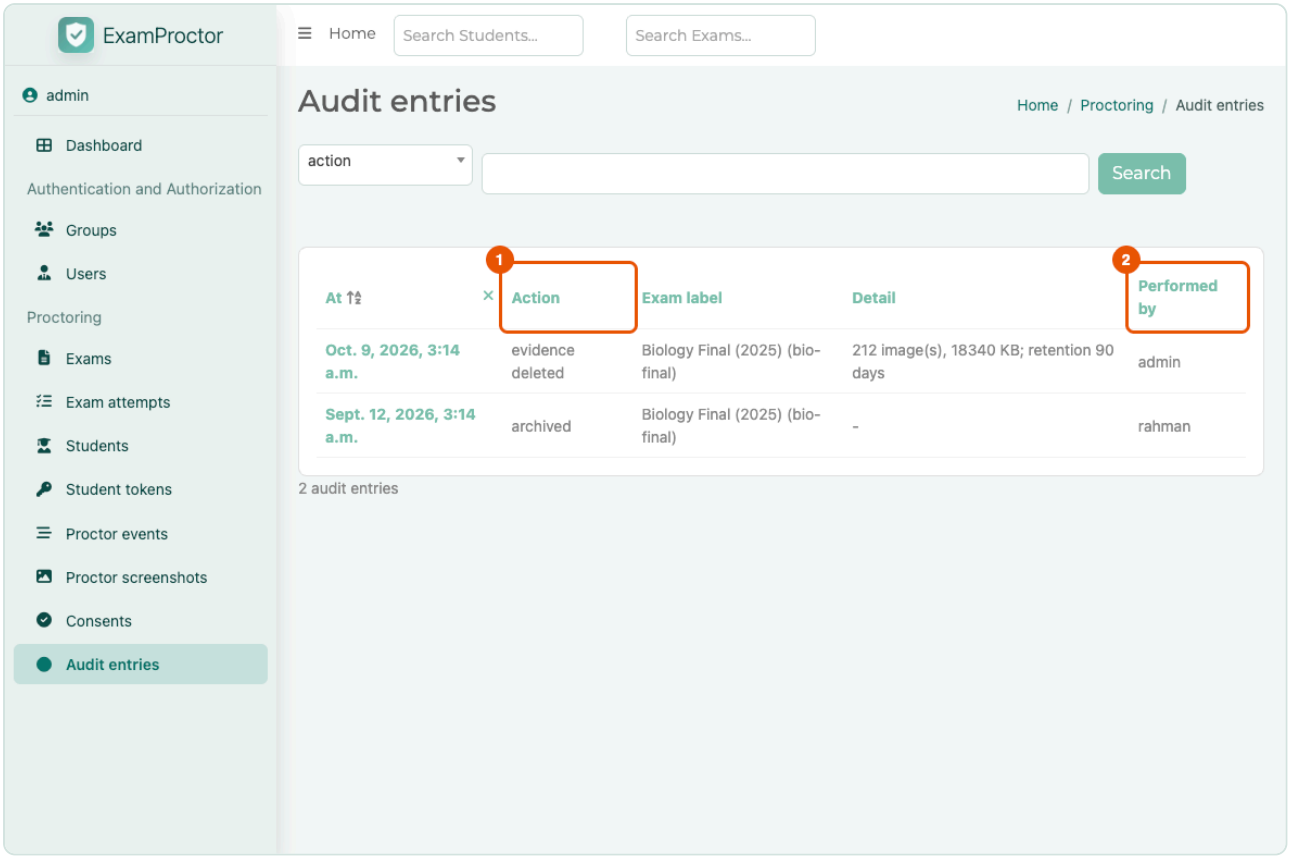
1 Choose the action, tick exams, press Go

- While the exam is still open, students in progress block archiving (end or finish them first).
- Once the exam time is over, attempts that never reported an ending (closed laptop, killed app) are completed automatically and the audit log notes it.

Delete evidence (screenshots and webcam photos)

- Open the exam's page in the teacher dashboard (`/teacher/exams/<id>/`), choose **Delete screenshots & photos**, type the exam id to confirm.
- Allowed only after the exam ended or was archived. Cannot be undone.
- Or automatically for old exams: `python manage.py purge_evidence --older-than-days 90 --dry-run` shows what would go; remove `--dry-run` to delete. Schedule it monthly if you like.

Audit entries



A permanent record of archives and deletions

1. What was done
2. Who did it

> **Why it matters**
> Audit entries cannot be edited or deleted, and they survive deleting the exam itself.

9 Server and maintenance

Settings (environment variables)

Variable	Purpose	Example
`DJANGO_SECRET_KEY`	Long random secret. **Required** in production.	(a long random string)
`DJANGO_DEBUG`	Set to `0` in production.	0
`DJANGO_ALLOWED_HOSTS`	Addresses the server answers to.	192.168.0.100

Variable	Purpose	Example
`TIME_ZONE`	Time zone for exam times and dashboards.	Asia/Dhaka
`SELF_ACTIVATION`	`1` = students can email themselves a password; `0` = hand-outs only.	1
`EMAIL_BACKEND`, `EMAIL_HOST`, `EMAIL_PORT`, `EMAIL_HOST_USER`, `EMAIL_HOST_PASSWORD`, `DEFAULT_FROM_EMAIL`	Email sending (Brevo or Resend over SMTP). Keep keys here, never in the code.	smtp-relay.brevo.com / 587
`STUDENT_DOWNLOAD_URL_MAC`	Link shown on the home page for the student installer.	(your Drive link)
`DATABASE_PATH`, `MEDIA_ROOT`	Where the database and uploaded images live.	/data/db.sqlite3
`ADMIN_API_KEY`	Optional: lets scripts change exam settings through the API.	(a secret)

Run `python manage.py check` after changing settings; it warns if email would only print to the terminal or if the development secret key is still in use.

Backups

Everything lives in two places. Copy both regularly (for example every night, and always before an update):

- the database file (`db.sqlite3` and its `-wal` companions while the server runs; stop the server for a perfectly clean copy)
- the `media/` folder (screenshots and webcam photos)

Updating the system

```
cd ExamApp && git pull
cd backend && source .venv/bin/activate
pip install -r requirements.txt
python manage.py migrate
# then restart the server
```

Performance and safety notes

- Twenty students logging in at once is fine. Password checking is deliberately slow, so use at least two server worker processes for large classes.

- Traffic on a school network is plain HTTP. That is acceptable on a trusted network; for anything wider, put the server behind HTTPS.
- If a reverse proxy sits in front of the server, make sure it passes the real client address, otherwise the per-address login limits would treat everyone as one.

Command reference

Command	What it does
`createsuperuser`	Create an administrator.
`import_students file.csv`	Add or update students from a CSV.
`create_teacher name --exams a,b`	Create or update a teacher and assign exams.
`deactivate_batch 81 [--dry-run]`	Deactivate every student of a batch.
`purge_evidence --older-than-days N [--dry-run]`	Delete old screenshots and photos.
`seed_exams`	Create two sample exams for a first test.
`check`	Health and safety checks of the settings.

10 Troubleshooting

What you see	Likely cause	What to do
Student sees no exams	Their batch differs from the exam's batch, the exam is archived, the exam has no batch, or the student has no batch.	Compare the batch numbers; set the exam's batch; move the student forward (or correct).
"Account not activated"	The student has no password yet.	Hand out a password (section 4) or let them activate by email.
"Too many failed attempts"	Five wrong tries from that computer.	Wait 15 minutes, or generate a new password.
"This exam has not started yet" / "has ended"	Outside the exam's start/end window (check the time zone!).	Edit the schedule, or give the student extra time.

What you see	Likely cause	What to do
Student cannot start: extra monitor / app names	The written part needs one screen and no screen-sharing, remote-access or recording tools (for example OBS, TeamViewer, Parsec, AnyDesk).	Disconnect the monitor / quit the app and try again. Practical exams allow a second monitor.
No screenshots appear	macOS Screen Recording permission is off for the app.	System Settings, Privacy & Security, Screen & System Audio Recording: turn on ExamProctor, restart the app. The monitor also shows a *Screenshot could not be taken* event.
Teacher cannot log in	Not in the Teachers group, locked out, or using the admin address.	Check the group; wait 15 minutes; use `/teacher/login/`.
Bangla names look garbled in Excel	Old exports without the encoding marker.	Current exports include it; re-download the CSV.
Emails never arrive	Email still set to print in the terminal, or SMTP details wrong.	Run `check`; set the `EMAIL_*` variables; try again (the student can retry safely).
Admin pages look unstyled	Static files not served (production).	Run `collectstatic` and serve static files, or use the development server.

11 Quick reference

Exam day in five lines

1. Students are in the system with the right **batch** and each has their **own password**.
2. The exam exists with the right **batch**, **teachers**, **schedule** and Google Form.
3. The server is running and awake; students have the installer with the correct server address.
4. Teachers watch the live monitor; you handle passwords, resets and server problems.
5. Afterwards: marks, archive, delete evidence after the retention period, deactivate the batch.

Rules to remember

- Every exam has exactly one batch. Batches move forward only (corrections are recorded).
- Written first, then practical. Quitting is final until reset.
- Passwords are shown once. Never share one password between students.
- Teachers use `/teacher/`, administrators use `/admin/`; neither links to the other.

- Deleting screenshots keeps the log and marks; deleting a student also deletes their attempts.